The `deciders` explanations reveal why the allocation fails. We can also fetch the details of a single unassigned shard using the same API by providing the index and the shard.

Listing 15.13 Getting an explanation for a single unassigned shard

```
GET _cluster/allocation/explain
{
  "index": "programming_books2",
  "shard": 0,
  "primary": true,
  "current_node": "mkmac.local"
}
```

This query fetches the allocation explanation for the given index. Based on the explanations, we can start troubleshooting to resolve unassigned shards and return the cluster to a GREEN state.

15.3.3 Disk-usage thresholds

Elasticsearch protects clusters from low disk memory by enabling disk-usage thresholds on the cluster. These thresholds are put in place to prevent nodes from running out of disk space and thus creating a failed cluster. Three thresholds are provided as low, high, and flood-stage disk watermarks for administrators to use accordingly.

LOW-DISK WATERMARK

The low-disk watermark is an 85% barrier threshold that kicks in if disk memory falls below 85% of total memory. If this threshold is crossed on a particular node, Elasticsearch does not allocate any new shards on that node until the node's memory drops below this setting. For example, if we have allocated 1 TB of disk memory to a node and 850 GB is already consumed, Elasticsearch raises this alert and takes measures to prevent node failures related to insufficient disk free space. The watermark can also be an absolute value, such as setting 200 GB instead of a memory threshold represented as a percentage.

Although this watermark threshold is set at 85% by default, we can use the cluster settings API to change it if needed. The following code shows the mechanism to change this. We decrease the low-disk-space watermark to 80%, which is a bit lower than the default.

Listing 15.14 Setting the low-disk watermark on the cluster

```
PUT _cluster/settings
{
  "transient": {
    "cluster.routing.allocation.disk.watermark.low":"80%"
  }
}
```